DEPARTMENT OF SOFTWARE ENGINEERING
BACHELORS IN SOFTWARE ENGINEERING
Course Code: SE-213
Course Title: Data Structures and Algorithms
Complex Engineering Activity
SE Batch 2024, Fall Semester 2025

CLO Covered:
CLO 4: (P3-PLO3)

Complex problem-solving attributes (CPA) covered (as per PEC - OBA manual – 2019)
CPA-1 Depth of analysis required: Problems require abstract thinking and analysis to design efficient algorithmic models.
CPA-2 Level of interaction: Require resolution of problems involving multiple data structures and algorithmic interactions.
CPA-3 Familiarity: Extend beyond previous experiences by applying principles-based approaches for optimization and analysis.

Complex Engineering Activity (CEA): ***Implementation of a Modern Instruction Set Architecture (ISA)***

Activity Overview

In this activity, students will design algorithms for a Smart Information Management System (SIMS) for a chosen context (e.g., student records, patient records, transport routes, or library system). The goal is to demonstrate understanding of data structures, algorithm design, and complexity analysis.

Implementation Guidelines

Select SIMS: Work on the Smart Information Management System (SIMS) submitted by you in CEP.

Design and Implement a Project: Create a simple application (e.g., a student record system, patient data tracker, or library management module) that effectively utilizes key **data structures** such as arrays, stacks, queues, trees, and graphs. Provide **algorithmic examples** that demonstrate core operations like searching, sorting, traversal, and data manipulation to showcase efficient data management and problem-solving techniques.

Document Your Implementation: Write a report detailing the implementation process, including:

- Overview of the chosen SIMS and the data structures used
- Explanation of algorithms implemented (insertion, deletion, traversal, sorting, etc.)
- Challenges faced during design or complexity optimization and how they were resolved

Deliverables: Implementation Code: Submit your code files by week 14.

Scoring Guide: Total Marks: 10.Each criterion is worth 2 marks, rated as follows:

- **0 = Unsatisfactory**
- **1 = Basic**
- **2 = Excellent**

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Name	
Roll Number	

Assessment Rubric

Criteria	CPA Mapping	0 (Unsatisfactory)	1 (Basic)	2 (Excellent)
Correctness of Implementation	**CPA-1: Depth of analysis**	*Code does not compile or runs incorrectly; no ISA features used*	*Code compiles but demonstrates basic functionality with minimal ISA usage*	*Code is fully functional and effectively demonstrates key ISA features*
Efficiency and Optimization	**CPA-1: Depth of analysis**	*No consideration for efficiency or optimization; poorly written code*	*Basic level of optimization; code is functional but could be improved*	*Highly efficient code that utilizes ISA features for optimal performance*
Code Clarity and Documentation	**CPA-2: Level of interaction**	*Code is poorly organized; lacks comments and documentation*	*Code is somewhat organized; minimal comments and documentation present*	*Well-organized code with clear and comprehensive comments explaining logic and design*
Real-World Application Relevance	**CPA-2: Level of interaction**	*No relevance to real-world applications; unclear use case*	*Some relevance to real-world applications; basic explanation provided*	*Strong relevance to real-world applications with a clear use case and impact*
Use of ISA Features	**CPA-3: Familiarity**	*No use of specific ISA features; irrelevant code*	*Limited use of ISA features; basic implementation of some instructions*	*Comprehensive use of ISA features, effectively demonstrating their advantages in the implementation*

Total	
Signature of Teacher	